

Sparse Table

Sparse Table is a data structure, that allows answering range queries. It can answer most range queries in $O(\log n)$, but its true power is answering range minimum queries (or equivalent range maximum queries). For those queries it can compute the answer in $O(1)$ time.

The only drawback of this data structure is, that it can only be used on *immutable* arrays. This means, that the array cannot be changed between two queries. If any element in the array changes, the complete data structure has to be recomputed.

Intuition

Any non-negative number can be uniquely represented as a sum of decreasing powers of two. This is just a variant of the binary representation of a number. E.g. $13 = (1101)_2 = 8 + 4 + 1$. For a number x there can be at most $\lceil \log_2 x \rceil$ summands.

By the same reasoning any interval can be uniquely represented as a union of intervals with lengths that are decreasing powers of two. E.g. $[2, 14] = [2, 9] \cup [10, 13] \cup [14, 14]$, where the complete interval has length 13, and the individual intervals have the lengths 8, 4 and 1 respectively. And also here the union consists of at most $\lceil \log_2(\text{length of interval}) \rceil$ many intervals.

The main idea behind Sparse Tables is to precompute all answers for range queries with power of two length. Afterwards a different range query can be answered by splitting the range into ranges with power of two lengths, looking up the precomputed answers, and combining them to receive a complete answer.

Precomputation

We will use a 2-dimensional array for storing the answers to the precomputed queries. $\text{st}[i][j]$ will store the answer for the range $[j, j + 2^i - 1]$ of length 2^i . The size of the 2-dimensional array will be $(K + 1) \times \text{MAXN}$, where MAXN is the biggest possible array length. K has to satisfy $K \geq \lfloor \log_2 \text{MAXN} \rfloor$, because $2^{\lfloor \log_2 \text{MAXN} \rfloor}$ is the biggest power of two range, that we have to support. For arrays with reasonable length ($\leq 10^7$ elements), $K = 25$ is a good value.